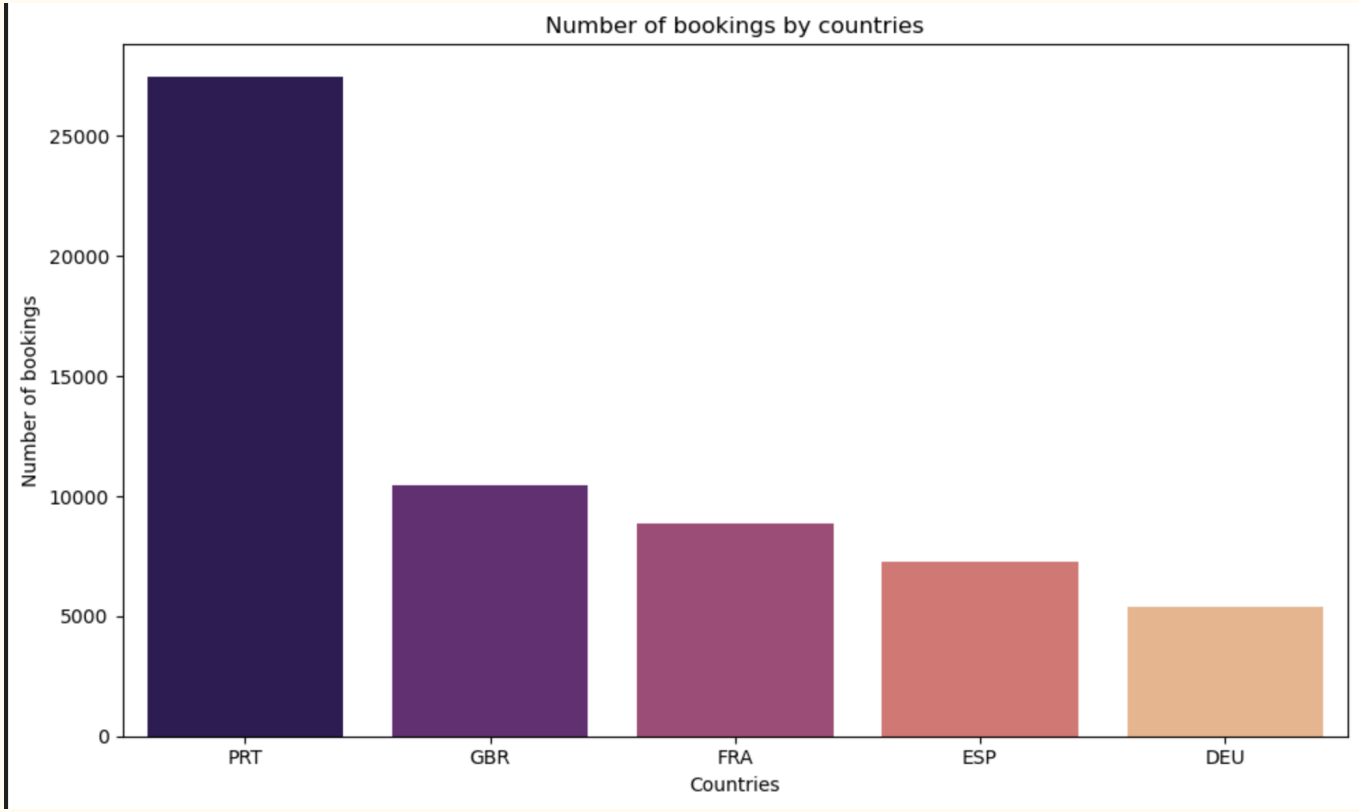
Predicting Hotel Booking Cancellations

Losel Lateng

Cleaning the data

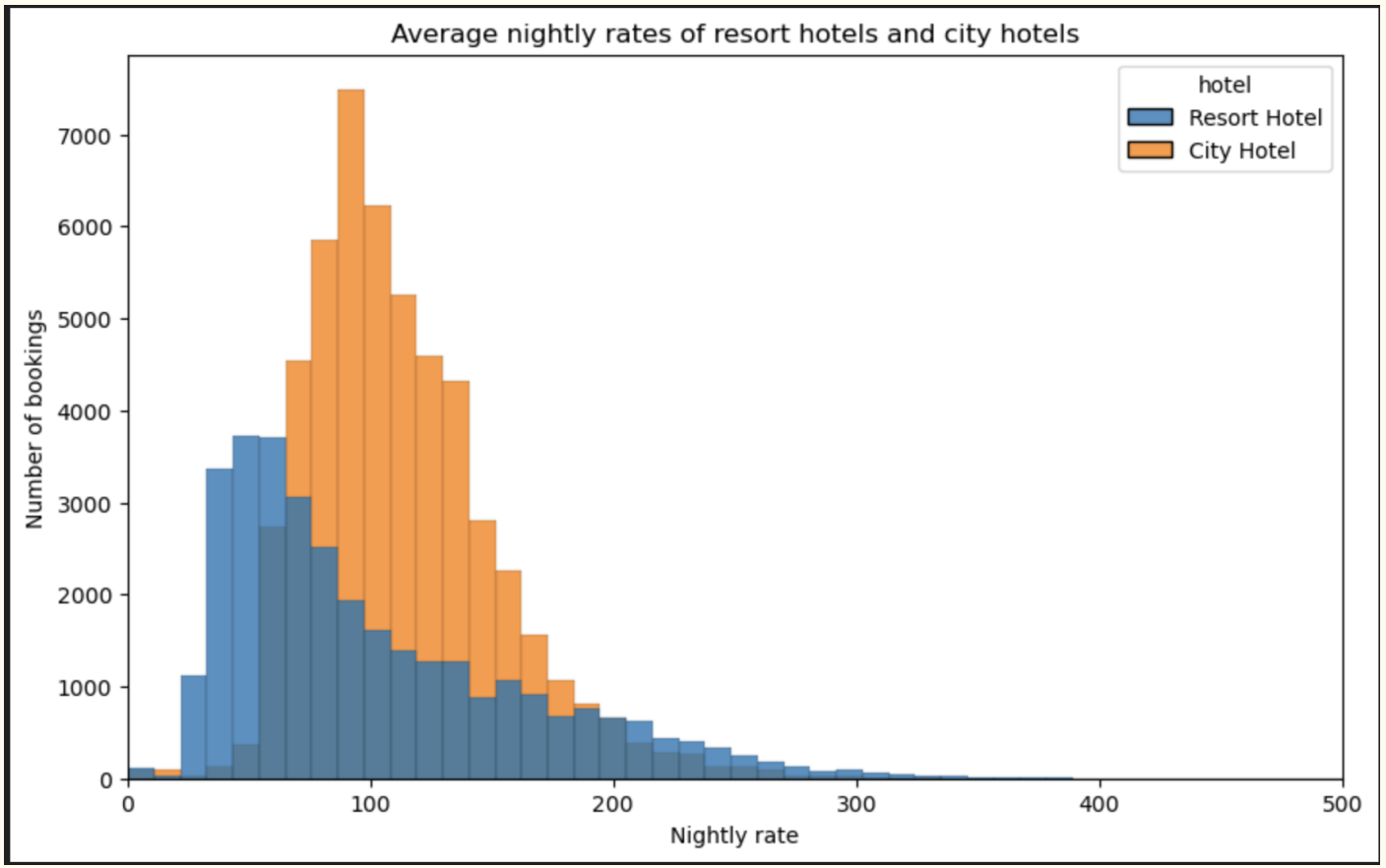
- Drops
  - Company-missing 94% entries
  - duplicates
  - reservation_status and reservation_status_date-hide answer from models
  - countryless entries-makes up small percentage of dataset, NaN or null values removed
- Filling missing/null values
  - Filling agent with 0 signifies the guest booked directly with the hotel if there is a missing value
  - Filling children with 0 because a missing value most likely means no children
  - Define agent and children as int types to correct NaN float type

Exploratory Data Analysis



PRT or Portugal is the country with the most hotel bookings.

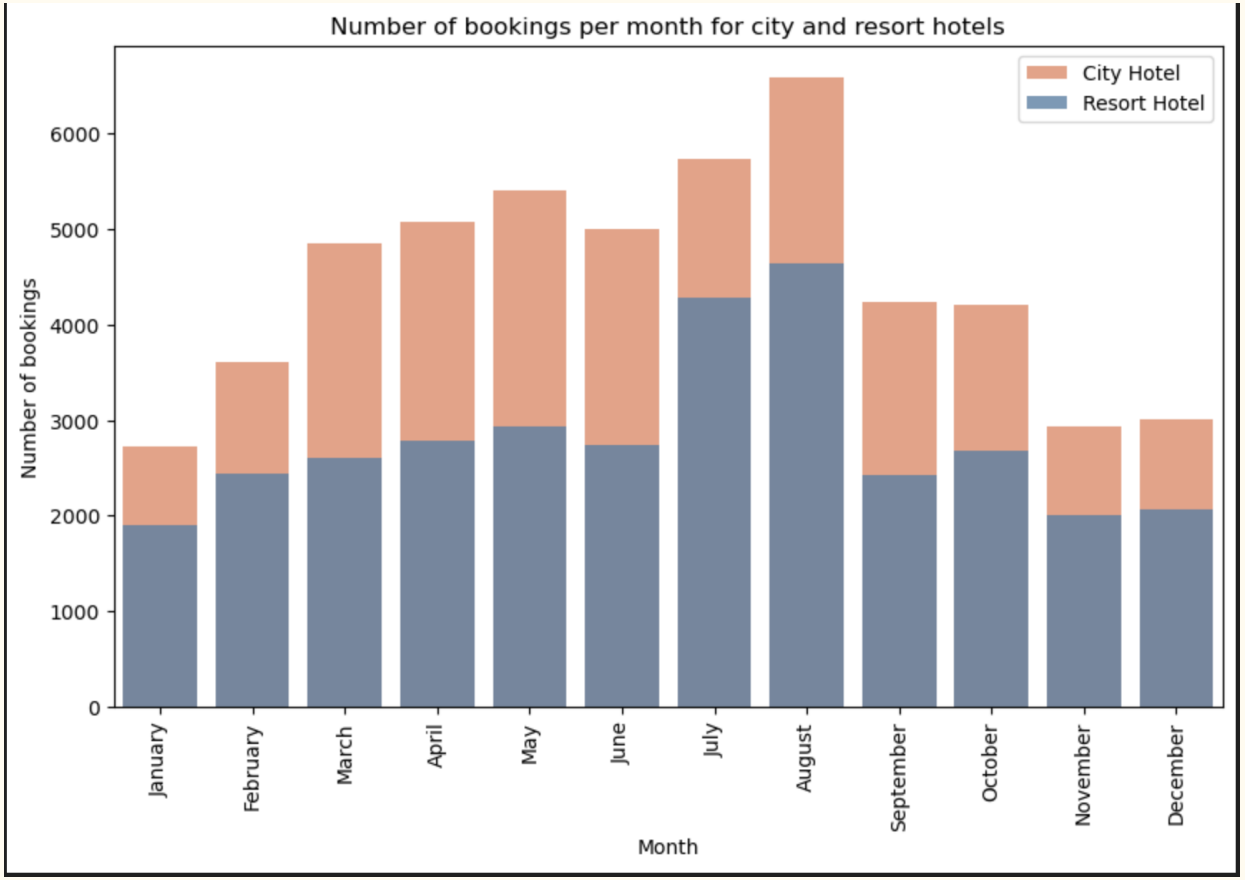
Exploratory Data Analysis



The city hotels have a much larger number of hotels compared to that of the resort hotels.

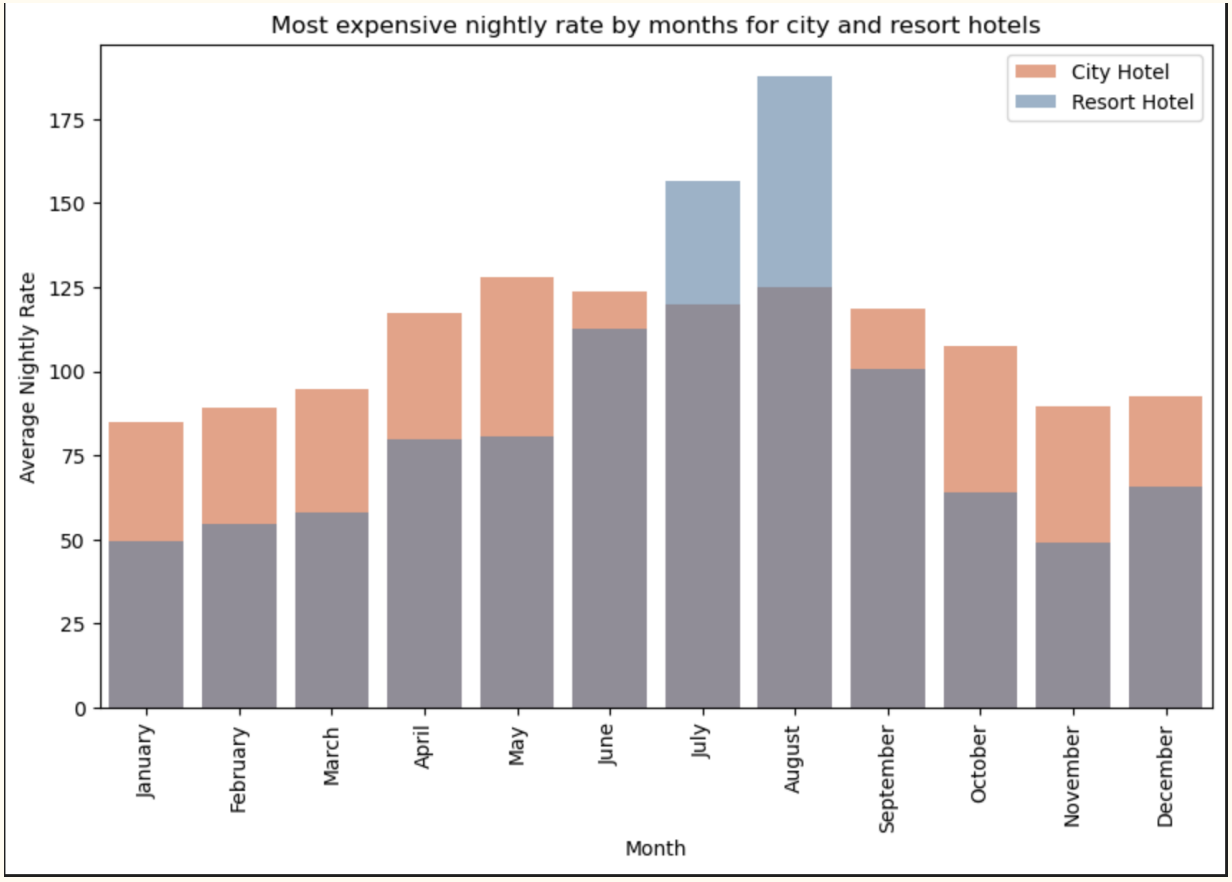
The resort hotels generally have a lower nightly rate compared to the city hotels.

Exploratory Data Analysis



The months that have the most number of bookings are August, July, and May.

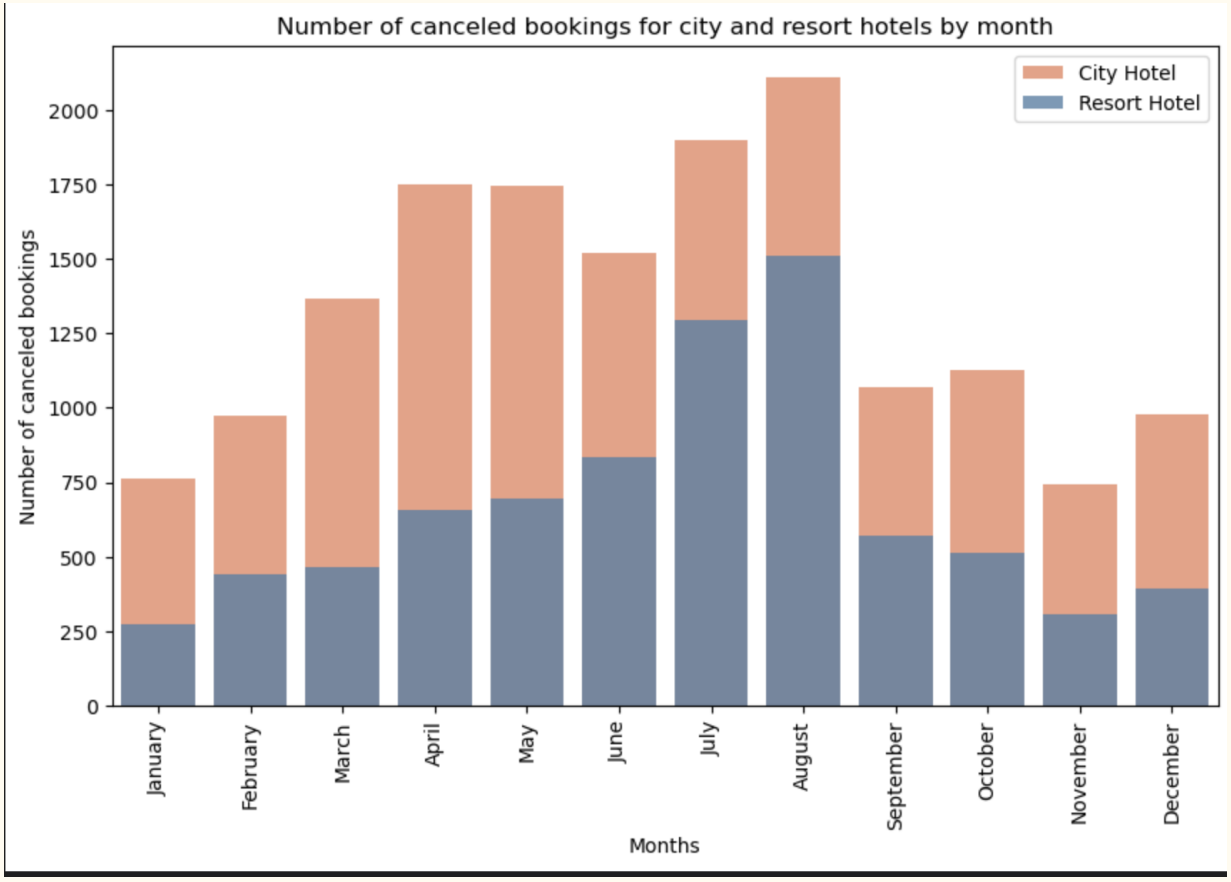
Exploratory Data Analysis



The most expensive nightly rates are in August, July, and May.

Resort Hotels have their third most expensive rates in July instead of May.

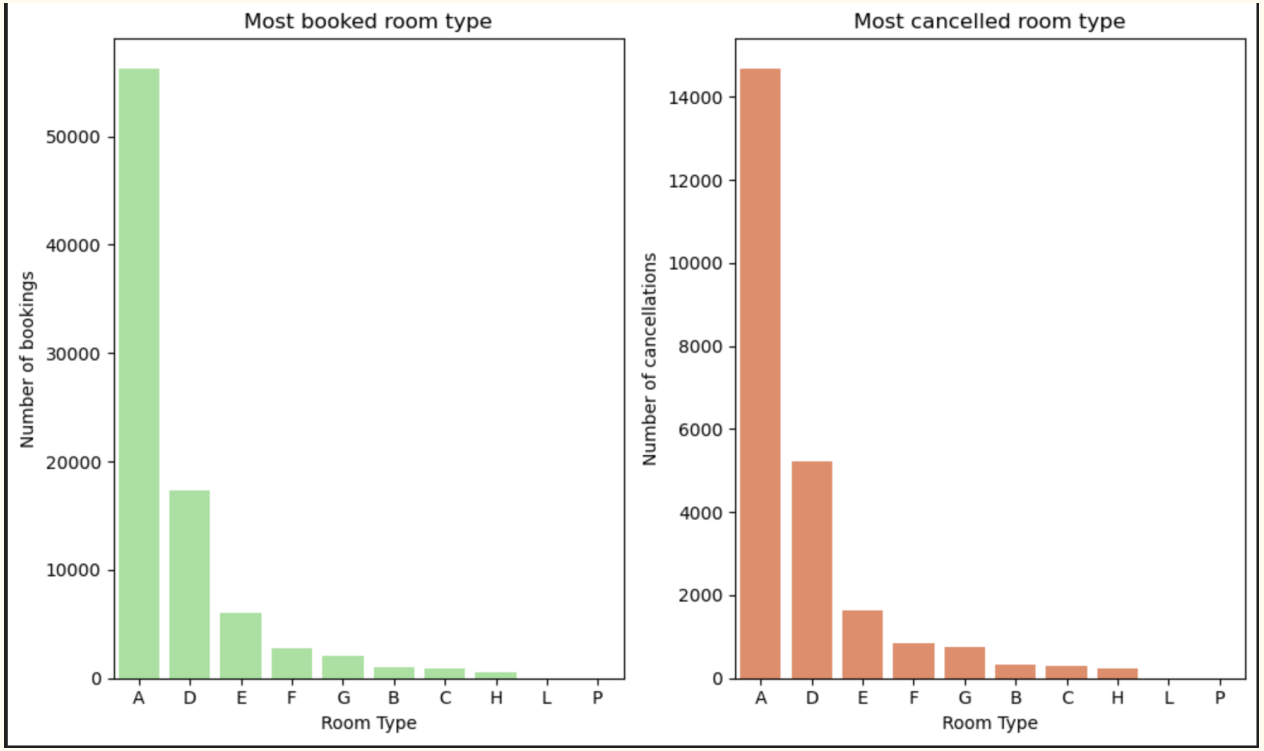
Exploratory Data Analysis



For both types of hotels, August and July are the most cancelled booking dates.

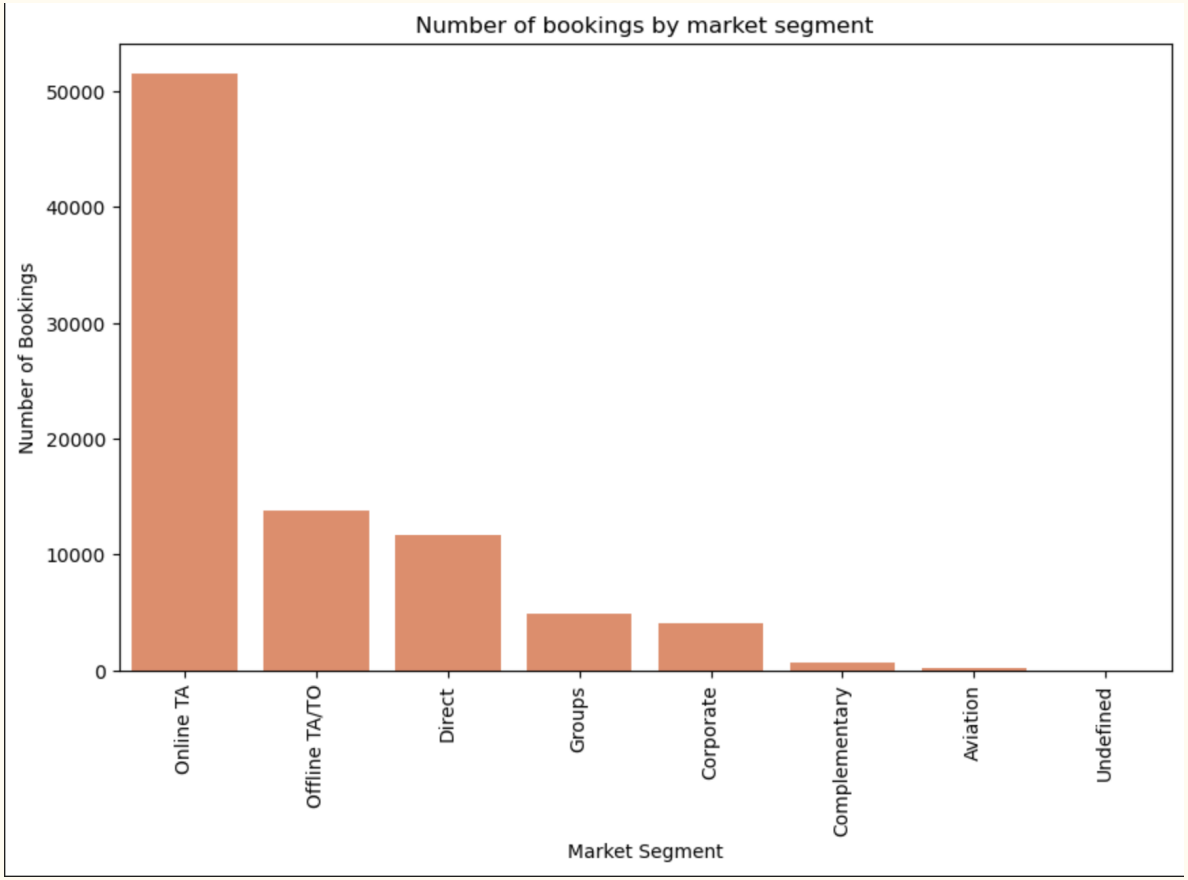
Followed by June for resorts and April for city hotels.

Exploratory Data Analysis



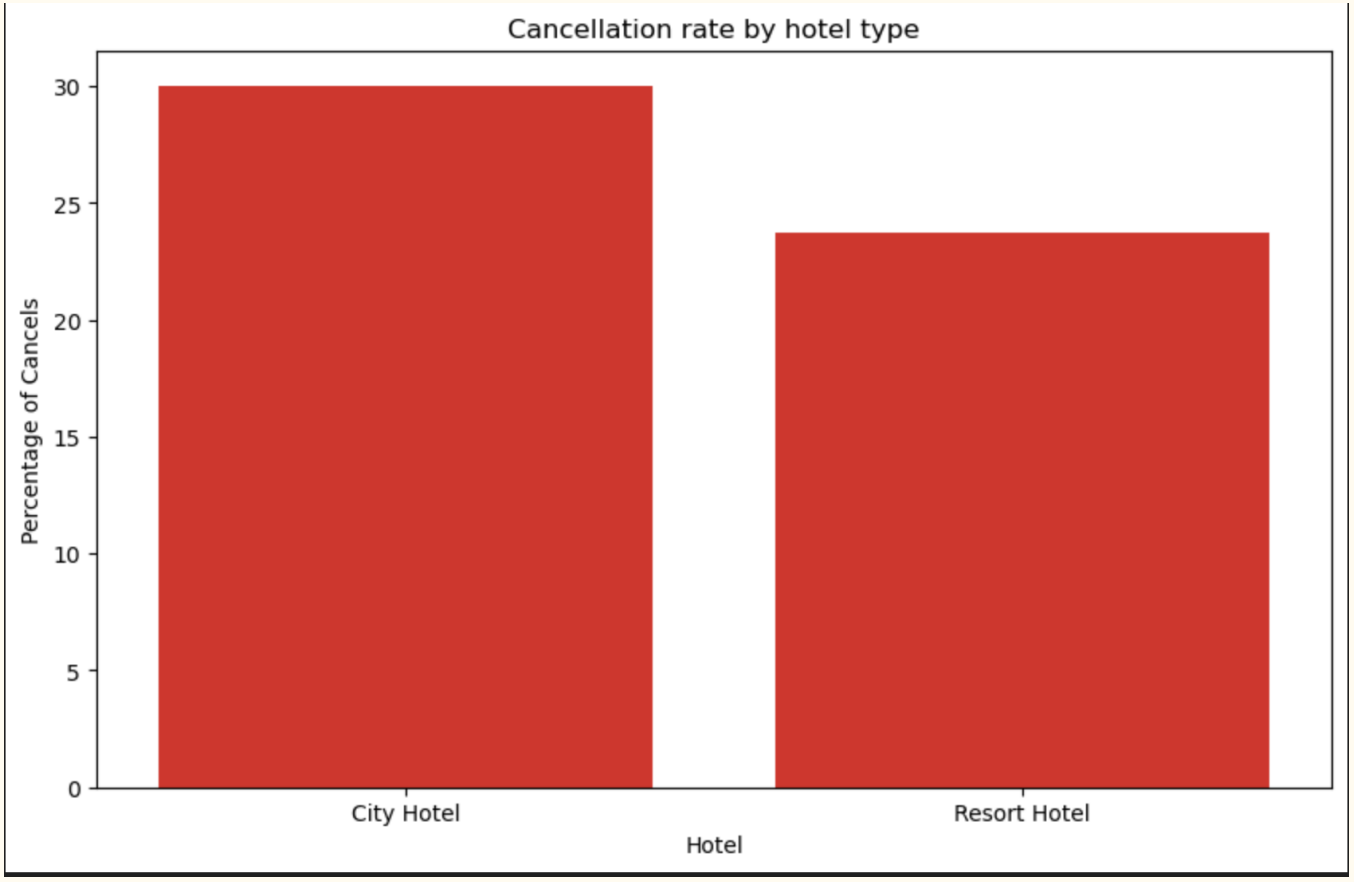
The most common rooms that were booked were rooms A, D, and E. These rooms also had the most cancellations.

Exploratory Data Analysis



The vast majority of bookings are done through online travel agencies.

Exploratory Data Analysis



City hotels see a larger cancellation rate compared to resorts.

Models

Logistic Regression

- Logistic Regression is a classification model that separates outcomes into a binary yes or no. In this case, outcomes are either canceled or not canceled. It determines the classification by a sigmoid function that converts the column features weighted sum into either a value of 0 or 1. The model threshold determines if the function's result should be cancelled or not cancelled.

Random Forest

- Random Forest is a model used for both regression and classification tasks that makes multiple decision trees during training. Each tree in the forest provides a prediction based on a random subset of features and data. For classification, the model takes the majority vote of all trees to determine the final category, while for regression, it averages the results. This model significantly improves predictive accuracy and prevents overfitting.

Decision Tree

- Used to model relationships between models and provide clear ways to make decisions. The decision tree could be used in this case to find out what data directly lead to a cancellation or not. I opted to use it for its ability to map out which specific variables impact booking cancellations.

Logistic Regression

- Dropped country before encoding and then one hot encoded the columns using pd.getDummies
- Is_canceled is the Y because we are predicting that
- Dropped is_canceled because that is the target to predict
- Used the Train_test_split function to make a training and testing set. 80% training and 20% testing.
- Scaled features to make sure large features don't dominate
- Used F1 score for Cross validation because there are imbalance and binary classes. (A stupid model with could guess with 72% accuracy on an imbalance dataset so its a bad metric to use for this dataset)

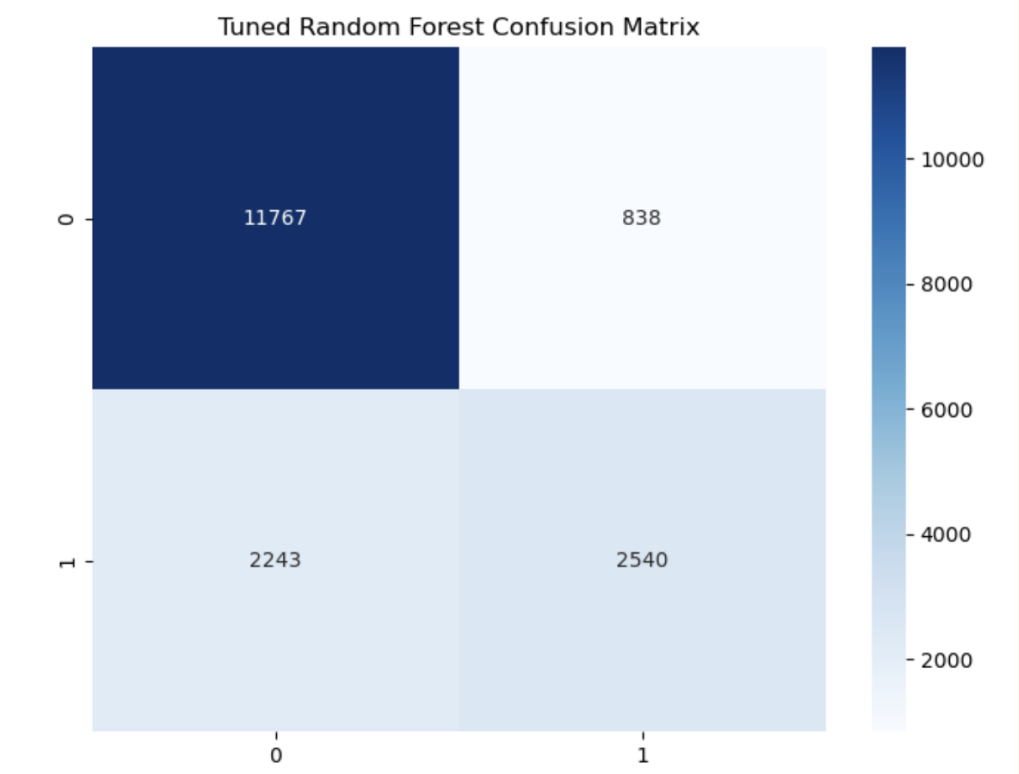
This was the baseline model. I wanted to test whether the relationship between cancellations and features was linear. Since the data had low accuracy, we can conclude that the data is more complex and not linear.

Logistic Regression P2

- We used the scikit library for logistic regression with 1000 iterations
- Implemented 5 fold cross validation and the CV scores were ~79% across the 5 folds.
- Accuracy: ~79%
- Precision: ~69%
- Recall: ~42%
- F1: ~52%

```
Logistic Regression: Cross-Validation Results
Accuracy for each of the 5 folds: [0.78971963 0.78691589 0.79410496 0.78806614 0.79466532]
Average Validation Accuracy:      0.7907
Accuracy:  0.7890
Precision: 0.6913
Recall:    0.4213
F1 Score:  0.5235
```

Random Forest



	precision	recall	f1-score	support
0	0.84	0.93	0.88	12605
1	0.75	0.53	0.62	4783
accuracy			0.82	17388
macro avg	0.80	0.73	0.75	17388
weighted avg	0.82	0.82	0.81	17388

Weighted average precision of 82%

Non-cancellations: precision of 0.84; recall of 0.93

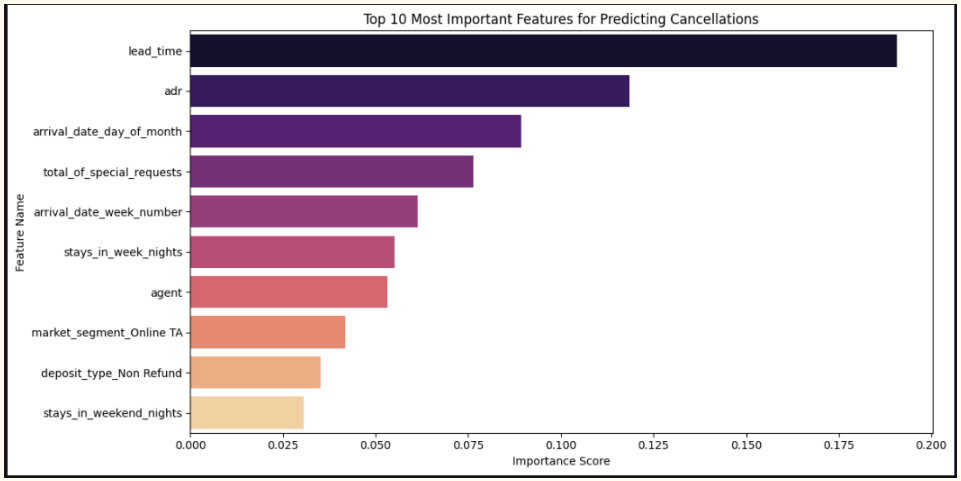
Actual cancellations: precision of 0.75; recall of 0.53

As seen by the confusion matrix, our model is strongest at predicting non-cancellations (11767 vs. 2540) due to non-cancellations severely outnumbering cancellations in our dataset.

Decision Tree

Metrics

- Stable model(75.67% average cross validation accuracy to 75.90% accuracy)
- Subpar Precision(55.99%), Recall(57.91%), F1 Score(56.94%)



Chart

- 10 most important features
- Lead_time has the highest importance score

```
Decision Tree: Cross-Validation Results
Accuracy for each of the 5 folds: [0.75514019 0.76175413 0.75981308 0.75298347 0.75361277]
Average Validation Accuracy:      0.7567
Accuracy: 0.7590
Precision: 0.5599
Recall: 0.5791
F1 Score: 0.5694
```

Conclusion

In this project, I explored hotel booking data to predict whether a customer would cancel their reservation. The first step was to clean the data by removing duplicate values, NaN, entries with missing values, null values, etc. Anything that would negatively impact the accuracy of the models. I also analyzed using a multitude of graphs comparing different data, including exploratory data analysis where we saw comparisons of the highest rates for different variables, the actual vs predicted values, as well as the most important features from the decision tree.

After cleaning and analysis, three models were created: Logistic Regression, Random Forest, and Decision Tree. Out of the three models, Random Forest had an accuracy of 82%, compared to Decision Tree at 75.9% and baseline model Logistic Regression. All three models struggled to predict actual cancellations due to an imbalance in the number of cancellations vs. non-cancellations.

Cross validation confirmed that our models generalized well but in the future, I could avoid this inaccuracy by balancing class weights so that the model penalizes more for missing cancellations.

Thank you

Any Questions?